



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Polynomials

POLYNOMIALS

"In scientific computing applications, mathematical polynomials are often represented using linked lists where each node stores a coefficient and exponent. Write a C program to represent two polynomials using linked lists and perform polynomial addition.

Test Case 1

Input

Polynomial 1: $5x^2 + 3x + 2$

Polynomial 2: $4x^2 + 6x + 1$

Output

Resultant Polynomial:

$9x^2 + 9x + 3$ "

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct Node {
5     int coeff;
6     int exp;
7     struct Node *next;
8 } Node;
9
10 Node* createNode(int coeff, int exp) {
11     Node *n = (Node*)malloc(sizeof(Node));
12     if (!n) {
13         printf("Memory allocation failed\n");
14         exit(1);
15     }
16     n->coeff = coeff;
17     n->exp = exp;
18     n->next = NULL;
19     return n;
20 }
21
```

OUTPUT

Enter number of terms in Polynomial 1: Enter terms (coeff exp) for Polynomial 1:

Enter number of terms in Polynomial 2: Enter terms (coeff exp) for Polynomial 2:

Resultant Polynomial:
2530x⁻⁵³⁵⁶⁸⁵¹³⁶

INPUT

$$5x^2+3x+2$$

$$4x^2+6x+1$$

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